# claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

## Metadata

```
- Source: `claude-windows`
- Kind: `claude`
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-inde
xer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_736be611-fb5\agent-
a128ad3b22e13bee4.jsonl`
- SHA-256: `3485a996e0409aee221fe4dd94cf2602df666687310731f8e75db07689a9d16a`
- Source modified: `2026-06-16T15:16:32+00:00`
- Imported at: `2026-07-05T16:48:27+00:00`
- project: `wf_736be611-fb5`
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

## Transcript

### 1. user (2026-06-16T15:14:30.302Z)

```
Adversarially VERIFY this finding from the real-footage rally-detection validation. Try hard to
REFUTE or show it is overstated, using ONLY the data (you MAY Read
output/local_vs_gemini_6.json). Default to holds=false if the numbers do not clearly support it.

CLAIM (trajectory-signal): mahadevpura-2 'fits' because its inter-rally dead time produces clean
inactivity gaps at roughly true-rally cadence, so the cap splits at real rally boundaries.
EVIDENCE CITED: mahadevpura-2 W1: n=40 vs Gemini 39, count_ratio 1.03, mean 13.8s, max 43.3s; on
the owner's 40-label ground truth the hard-cap variant reaches F1@0.25 0.722 and mean-IoU 0.50 ?
by far the best local fit, vs baseline 0.000/0.03.


REAL-FOOTAGE VALIDATION DATA (6 Kushagra/Yash June-12 clips, 1080p proxies, fully-LOCAL no-AI
WASB
windowing vs Gemini. Gemini = strong REFERENCE, NOT ground truth ? these 6 have no golden labels
EXCEPT mahadevpura-2 which the owner hand-labeled, see GROUND-TRUTH below).

Rally COUNT (Gemini | local baseline | local W1 cap=12 | local W1+W2 mc=1):
  GX010128: 39 | 15 | 78 | 75
  GX010129: 27 | 2  | 50 | 47
  GX020125: 11 | 6  | 18 | 15
  mahadevpura-0: 40 | 19 | 62 | 43
  mahadevpura-1: 9 | 1 | 2 | 2
  mahadevpura-2: 39 | 5 | 40 | 40
  AGGREGATE: Gemini 165 | baseline 48 (mean window 62.9s, 27 merge-groups, mIoU 0.23) |
             W1 250 (mean 10.8s, 10 merge-groups, mIoU 0.39) |
             W1+W2 222 (mean 11.5s, 10 merge-groups, mIoU 0.39, 6 gemini-only "misses")
  Per-video mean window (W1): GX010128 7.8s, GX010129 9.3s, GX020125 14.1s, mahadevpura-0 10.4s,
             mahadevpura-1 86.9s (the continuously-active blob W1 cannot cap ? no inactivity
gap),
             mahadevpura-2 13.8s. Gemini means ~5.7-7.2s.

GROUND-TRUTH (mahadevpura-2, 40 owner-labeled rallies, mean 5.7s, verified frame-aligned):
  F1@0.5 / F1@0.25 / mean-IoU:  Gemini 0.557/0.835/0.55 ; local W1+W2 0.150/0.500/0.31 ;
  local W1+W2+hard-cap 0.296/0.722/0.50 (best local). baseline 0.000/0.000/0.03.
  BOUNDARY ERROR (matched rallies, seconds): Gemini |?start| med 1.46 |?end| med 1.19 ;
  local-hardcap |?start| med 1.74 (~= Gemini!) |?end| med 3.12 (>> Gemini).
  => local's rally-START detection is ~at par with Gemini; the gap is the rally-END.
  IoU<->time: for a 5.7s rally, tIoU 0.5 tolerates only ~+-1.9s shift (the ~1-1.5s service
window
  nearly exhausts it -> tIoU 0.5 over-penalizes; owner wants a +-2s tolerance metric).

PRIOR GOLDEN DELTAS (n=6, then n=7 incl. mahadevpura-2): baseline F1 0.300 -> W1 0.439 (+0.139,
6/0)
  -> W1+W2 0.457 (+0.019, 6/0); hard-cap golden-neutral on n=6 (+0.006 n.s.) but became BEST on
```

n=7
  (continuously-active clips). All flags default-OFF, promoted only on measured LOVO lift +
sign-test.

WINDOWING KNOBS: a frame is "active" iff shuttle visible AND per-second-normalized velocity >
  velocity_thresh(0.02); active frames within merge_gap(2.0s) coalesce; min_window(2.0s). W1 =
  max_window_duration cap(12s) split at the largest internal inactivity gap. W1.5 hard-cap =
  midpoint-chop when no gap. W2 = net-crossings gate (drop windows with <1 net crossing).

### 2. assistant (2026-06-16T15:14:33.620Z)

I'll verify this claim by examining the actual data file and checking the cited numbers against
it.

### 3. user (2026-06-16T15:14:34.459Z)

File does not exist. Note: your current working directory is C:\Users\avidu\Projects\badminton-
highlight-indexer\.claude\worktrees\w2-safeguard.

### 4. assistant (2026-06-16T15:14:37.071Z)

The file isn't at the expected path. Let me locate it.

### 5. user (2026-06-16T15:14:37.511Z)

No files found

### 6. user (2026-06-16T15:14:40.040Z)

eval_baselines\gemini_wrapper_2026-06-10.json

### 7. assistant (2026-06-16T15:14:44.598Z)

The cited data file `output/local_vs_gemini_6.json` does not exist in this worktree. Let me
search more broadly for any file containing the relevant data (mahadevpura-2, etc.) so I can
verify against actual numbers rather than just the prompt's restated figures.

### 8. user (2026-06-16T15:14:45.459Z)

Found 6 files
docs\RALLY_QUALITY_RESEARCH.md
docs\QUALITY_ITERATIONS.md
backend\pipeline\segmenters\fusion_hybrid.py
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\heuristic_lovo_n6.json
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md

### 9. user (2026-06-16T15:14:48.613Z)

C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-72-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-73-### 2.1 L1 (shuttle
tracking) is NOT the bottleneck
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-74-On the divergence clips
WASB reported the shuttle visible in **96?99 % of frames** (mahadevpura-1
docs\RALLY_QUALITY_RESEARCH.md:75:98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory
signal is present and dense.
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-76-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-77-### 2.2 L2

(segmentation) is the bottleneck ? and the flaw is conceptual
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\RALLY_QUALITY_RESEARCH.md-78-The windowing is `trajectory_to_action_windows`

### 10. assistant (2026-06-16T15:14:51.273Z)

Let me read the relevant sections of the research docs to find the actual recorded numbers for mahadevpura-2, the W1 cap, and the hard-cap variant.

### 11. user (2026-06-16T15:14:52.490Z)

1 # Rally-Detection Quality: Research, Eval Framework & Project Plan
2
3 **Status:** Foundation doc (2026-06-16), run as a **tracked project** (§7) ? *"Bridge the
4 local?Gemini rally-detection gap,"* **completed when all committed tiers land and a single quality
5 number is attributed (§7.1).** The empirical + research backbone for closing the gap between the
6 **local, no-cloud rally segmenter** and the **Gemini** path. Iterations land as their own measured
7 PRs, logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md); this doc is the plan they execute against.
8
9 **How to read this:** §1 frames the gap with measured numbers. §2 is the precise, code-grounded
10 diagnosis (read before proposing fixes). §3 inventories the eval/experiment harness we already
11 have. §4 specifies how we strengthen that harness (the "really strong evaluation framework").
12 §5 is the cited external-research synthesis (4 dimensions). §6 is the prioritized roadmap with
13 falsifiable success criteria. **§7 is the project plan ? work-item tracker, sequencing, the single
14 number we attribute to the effort, and the definition of done.** §8 recaps the non-negotiable
15 constraints + cross-links.
16
17 > **This doc does NOT duplicate prior design.** It builds on, and cross-links:
18 > [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) (the 2-layer mental model),
19 > [`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) (the A/B + LOVO harness, P1 shipped),
20 > [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) (shuttle+person+audio fusion),
21 > [`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) (the signal-fusion framework +
22 > §13 literature review ? **already** establishes that our skeleton is standard TAL/TAS +
23 > late-fusion; we *cite, don't re-derive*), [`ABLATION_LAYER.md`](ABLATION_LAYER.md),
24 > [`EVALUATION.md`](EVALUATION.md), [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)
25 > (TAL-head-first ? VLM), and [`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) (corpus growth).
26
27 ---
28
29 ## 1. The gap (measured)
30
31 Two paths produce rally segments today. **Gemini** (cloud VLM, watches the video semantically)
32 is strong; the **local** path (WASB shuttle trajectory ? velocity windowing, no cloud) lags
33 badly. Measured F1 vs golden labels ([`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md)):
34

| Path | Mahadevpura (clean, single court) | Kushagra (multi-court) |
|---|---|---|
| Heuristic velocity windowing (local today) | 0.657 | **0.157** |
| Gen-0 owned head (LogReg on 5 trajectory features, LOVO n=6) | 0.744 | 0.561 |
| Two-stage WASB + Gemini refine | 0.83 | ? |
| **Gemini wrapper (gemini-flash)** | **0.93** | **0.66** |

The local heuristic is **footage-fragile**: fine on a clean single court, collapses on multi-court / busy venues. The owned Gen-0 head already closes most of the multi-court gap (0.561 vs 0.66) ? the residual is **corpus-sized, not architecture-sized**.

### 1.1 The 2026-06-16 local-vs-Gemini divergence run (no golden labels)

A fully-local run (`wasb_hybrid --skip-ai-handoff`) vs Gemini on 3 fresh matches (Kushagra/Yash June-12; report: `?/June 12 Local-vs-Gemini/_LOCAL_vs_GEMINI_REPORT.md`, generator `scratch/compare_local_vs_gemini.py`). **Gemini is a strong reference here, NOT ground truth** (these clips are unlabeled) ? so this is an agreement/divergence study:

| | Gemini | Local (WASB no-AI) |
|---|---|---|
| rallies / windows (3 videos) | 59 | 12 |
| total "play" | 6:28 | **17:41** |
| mean window duration | **6.1 s** | **65 s** (10.6×) |
| true misses (gemini-only) | ? | **0** |
| merge groups (one local window ? ?2 Gemini rallies) | ? | 7 |

Headline: **the local path does not *miss* rallies ? it fails to *segment* them.** One mahadevpura-1 window spanned the entire 174 s clip. Local "play" is 2.7× Gemini's because its mega-windows swallow the dead time between rallies.

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## 2. Diagnosis (code-grounded ? read before proposing fixes)

The local pipeline is **two layers** (see [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md)): **(L1) detection** ? per-frame shuttle position (WASB/TrackNet); and **(L2) segmentation** ? turn the trajectory into rally intervals. The gap is almost entirely **L2**.

### 2.1 L1 (shuttle tracking) is NOT the bottleneck
On the divergence clips WASB reported the shuttle visible in **96?99 % of frames** (mahadevpura-1 98.6 %, GX020125 72.8 %, mahadevpura-2 78.8 %). The trajectory signal is present and dense.

### 2.2 L2 (segmentation) is the bottleneck ? and the flaw is conceptual
The windowing is `trajectory_to_action_windows` ([`backend/pipeline/detectors/tracknet_runner.py:256`](../backend/pipeline/detectors/tracknet_runner.py)):

- A frame is **"active"** iff the shuttle is **visible** AND its **per-second-normalized speed** exceeds `velocity_threshold` (default 0.02). Velocity = `(dist / frame_width) * (fps / dframes)` ? **already fps-normalized** (? fraction of frame-width per second).
- Active frames within `dead_time_merge_gap` (2.0 s) are **merged** into one window.
- Windows shorter than `min_action_window_duration` (2.0 s) are dropped.

**The conceptual flaw: "the shuttle is moving" is treated as "a rally is in progress."** During dead time the shuttle is still visible and gently moving (being picked up, walked back, knock-ups, slow serve prep) ? those frames read as *active*. The 2 s merge then bridges the brief gaps, and because there is **no `max_window_duration` cap and no inactivity/boundary split**, a single window grows until the shuttle finally goes invisible ? which, at 96?99 % visibility, is rarely. Result: rally + dead-time + rally + ? collapse into a few mega-windows.

> **Correction to an earlier hypothesis (honesty note).** A first pass guessed the cause was "velocity threshold not fps-invariant." It IS fps-invariant (the `fps/dframes` factor). The real cause is the *motion-equals-rally* conflation + 2 s bridging + no upper bound. fps matters only secondarily: denser 60 fps sampling keeps more slow-motion dead-time frames above the speed threshold. The `_LOCAL_vs_GEMINI_REPORT.md` has been corrected to match.

### 2.3 What this implies
The local path **has no rally-STATE signal** ? only "is the shuttle moving." Bridging the gap is fundamentally about giving L2 signals that separate *rally* from *shuttle-merely-moving* (net-crossings, two-sided play, sustained fast exchange, hit sounds) and a segmenter that produces *bounded, well-cut* intervals ? exactly the multi-signal-fusion + learned-segmenter direction the existing docs lay out. This doc makes that concrete and measurable.

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## 3. What we already have (eval/experiment harness inventory)

The harness is rich and composable ? we **extend** it, not reinvent it. (Modules under `backend/eval/`.)

**Scoring & matching**
- `metrics.py` ? temporal-IoU interval matching; precision/recall/F1, mean_iou, boundary errors.
- `segmentation_metrics.py` ? **F1@{0.1,0.25,0.5}**, **mAP@tIoU** (0.3?0.7), **segment_count_ratio** (the over/under-segmentation diagnostic). (Edit/segmental score deliberately omitted to date.)

**Statistical rigor (small-n)**
- `eval_stats.py` ? bootstrap 95 % CIs, paired exact **sign-test**, `paired_delta_ci`, `grouped_folds` (LOGO), `out_of_fold_predictions` (anti-stacking-leakage). The promotion bar = "?F1 CI excludes 0 **AND** sign-test agrees."
- `score_calibration.py` ? Platt + isotonic calibrators, **Brier + ECE**.

**Variants, A/B, leaderboard**
- `experiment.py` ? `Variant` (declarative recipe: segmenter + config + cue_flags + gates + learned-classifier + threshold/merge_gap/calibrate), `compare()` (LOVO-honest A/B with an "honest" stats block), `compare_unlabeled()` (agreement on unlabeled footage), `CONTROL` (pinned baseline), `record_experiment()` ? `eval_baselines/experiment_leaderboard.json`.
- `ablation.py` ? leave-one-signal-out marginal ?F1 with CI + sign-test + a no-over-claim verdict vocabulary (`earns_keep` / `suggestive` / `inert` / `structural_protected`).
- `classifier.py` ? dependency-free numpy `LogisticRegression` (the learned scorer; JSON-serializable).

**Eval?serve discipline + gates**
- `windowing.py` ? single-sources the **SERVED** preset from `IndexingConfig` defaults so eval
  and production can't silently diverge; `PROPOSAL` (recall-oriented) is where the golden feature
  substrate is built. (This exists *because* Gate-A measured **+0.074 on PROPOSAL but ?0.008 on
  SERVED** and was reverted ? the canonical eval?serve lesson.)
- `promotion.py` / `served_gate_a.py` ? promote a cue only if it does NOT regress at the served
  windowing; `per_user_gate.py` / `per_user_regression.py` ? per-user analogues; `regression.py` ?
  golden-set no-regression baseline with a GT-hash to detect label changes.

**Golden substrate**
- `gt_loader.py` ? one canonical GT loader (both legacy formats). `fusion_golden.py` /
  `fusion_compare.py` / `fusion_audio.py` ? extract replayable per-window feature JSONs
  (shuttle 5 + person 5 + audio 3) so ablation/compare re-score on CPU without re-detecting.
- **Corpus today: 6 golden videos** (label-set hash `50d98ffbc370`); ~10 more expected via the
  Roora vendoring track ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)).

**Signals available to L2 (built, default-OFF):** net-crossing count (`rally_gate.py`, Gate-A),
5 person/court features (`person_detector.py` + `fusion_features.py`), 3 audio-hit features
(`audio_features.py`). Fusion hooks: `fusion_hybrid.py::_enrich_candidate_stats` /
`_select_for_handoff`. Persisted in `candidate_segments.stats`.

### 3.1 Known gaps in the harness (what §4 fills)
1. **Over-segmentation is under-surfaced in the default report.** `segment_count_ratio` exists but
   the merge/split *breakdown* (the very thing that exposed the over-merge) lives only in a scratch
   script. tIoU-F1 alone is **blind to over-merge** ? a giant window can still "match" via overlap.
2. **No segmental edit-score** (Levenshtein over the predicted/gt label sequence) ? the standard
   TAS over-segmentation metric.
3. **Per-cue calibration is fitted but never *measured* post-hoc** (no ECE/Brier delta reported).
4. **`out_of_fold_predictions` (anti-leakage stacking) is built but not wired to an arbiter.**
5. **No active-learning loop** ? nothing tells us *which* unlabeled video to label next.
6. **No single "rally-segmentation eval" entrypoint** ? the pieces exist but a one-command
   `score this segmenter vs golden, honestly` runner would make every iteration cheap.
7. **The local-vs-Gemini agreement track is ad-hoc** (a scratch script), not a first-class
   *shadow* signal alongside the golden gate.

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## 4. The strengthened evaluation framework (the "really strong harness")

Design principles (inherited, non-negotiable):
- **Signals ? learned scorer, NO special-casing** ([`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) §1):
  every new cue is a feature column the scorer learns to weight; never an `if/else` in the decision path.
- **Promote only on measured lift; default OFF; revert is one flag** ([`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) §6).
- **eval ? serve:** a harness win must be re-confirmed on the *served* windowing before

any default flip.
- **Honest small-n stats:** never a bare mean at n?6?16; always CI + paired sign-test (C1).

### 4.1 Metric set the leaderboard should standardize on
Surface ALL of these in the default `compare()` report (most already computed; the task is to
*report* them together so over-merge can never hide again):

- **tIoU-F1 @0.5** (primary) **and F1@{0.1,0.25,0.5}** (segmental, TAS-standard).
- **mAP@tIoU** (0.3?0.7) when predictions carry confidences.
- **`segment_count_ratio`** + a **merge-rate / split-rate** breakdown (fraction of GT rallies that
  share a predicted window = under-segmentation; fraction of predicted windows that span ?2 GT =
  the over-merge signal). Promote the scratch `compare_local_vs_gemini.py` matching into a reusable
  `backend/eval/` helper.
- **Segmental edit (Levenshtein) score** ? the canonical over-segmentation metric we currently lack.
- **Boundary timing** (mean start/end error on matched pairs) ? already in `metrics.py`, surface it.
- **Calibration (ECE/Brier)** reported pre/post per-cue calibration (close gap #3).

> **Why:** plain tIoU-F1 rewarded our mega-windows (they "overlap" real rallies). The metric set
> above makes over/under-segmentation *first-class*, so a fix that improves segmentation is
> visible even when tIoU-F1 barely moves ? and vice-versa.

### 4.2 Splits & significance
- **Leave-one-video-out, grouped by match** (never leave-one-*window*-out ? windows within a video
  are correlated). Keep `grouped_folds`. Note the **RLOOCV bias** caveat for tiny n (see §5.4).
- **Bootstrap CI + paired exact sign-test** on per-video held-out F1 (C1). A result is *promotable*
  only if CI excludes 0 **and** the sign-test agrees, **and** it survives the served-windowing check.
- **Anti-leakage:** per-LOVO-fold threshold tuning + calibration on the TRAIN fold only (already
  enforced in `experiment._calibrate_and_tune`); wire `out_of_fold_predictions` once an arbiter exists.

### 4.3 The golden corpus + active learning (the highest-leverage lever)
The single biggest quality lever is **more labeled matches** ([`NEXT_STEPS.md`](NEXT_STEPS.md):
"corpus growth = highest leverage"). With ~6 ? ~16 videos:
- Grow via the Roora vendoring flow ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md),
  [`ROORA_LABELING_GUIDELINES.md`](ROORA_LABELING_GUIDELINES.md)) ? court calibration, IAA, the
  3 non-negotiables (ignore dead time, every rally, one-contact = one shot).
- **Active learning:** stop labeling videos at random. Score the *unlabeled* corpus and label the
  videos where the local segmenter is most *uncertain* / most *divergent from Gemini* / most
  *diverse* (new venue, fps, lighting). §5.3 surveys the methods; the local-vs-Gemini agreement
  signal (§4.4) is a ready proxy for "where do we disagree most ? label that."

### 4.4 Shadow evaluation: Gemini-agreement as a free, unlimited signal (NOT a gate, NOT training data)
We have exactly one strong reference available on *unlabeled* footage at inference time: **Gemini**.

- `experiment.compare_unlabeled()` already scores agreement (agreed / a-only / b-only) without labels.
- Use it as a **shadow metric**: run any candidate segmenter on a large unlabeled pool, measure agreement-with-Gemini, and watch it as a *trend* ? cheap, unlimited, catches regressions the 6 golden videos can't. **It is a shadow signal only.** Golden labels remain the gate; Gemini is not ground truth (it has its own errors) and ? per the licensing guardrail (§8) ? **its outputs may never become training data for owned weights.**

### 4.5 One-command rally-segmentation eval (close gap #6)
Wrap the above into a single reusable entrypoint (a future PR ? not built here): given a segmenter variant + the golden manifest, emit the full §4.1 metric set + the honest stats block + a leaderboard row, and (optionally) the §4.4 shadow agreement on an unlabeled pool. Every iteration in §6 is then "run the entrypoint, read the honest delta, promote-or-not."

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## 5. External research synthesis (cited, adversarially verified)

> Produced by a deep-research run (6 search angles ? 27 sources ? 126 candidate claims ? 25 3-vote-adversarially-verified, 23 confirmed / 2 refuted). **Cite, don't over-claim.** Two claims were *killed* in verification ? recorded in §5.5 so we don't repeat them. **License caveat:** the code repos below are research releases; **independently confirm each repo's actual license is MIT/Apache-2.0/BSD before vendoring any of it** (this survey did not verify every license file).
> **Governance:** none of these techniques require training on paid-LLM outputs ? all are compatible with our no-Gemini-training-data guardrail (§8).

### 5.1 Windowing & boundaries (TAL/TAS) ? the explicit-boundary machinery we lack
The recurring lesson: modern action-localization models have an **explicit boundary mechanism**; our heuristic has none (it merges on a velocity+gap rule). The transferable *idea* is "predict a boundary and cut there," but the deep models are built for **hundreds+ of training videos**, so at n=6?16 they are **research bets**, not drop-ins.

- **ASRF ? Action Segment Refinement Framework** (Ishikawa et al., WACV 2021, [arXiv:2007.06866](https://arxiv.org/abs/2007.06866)).
  A shared feature extractor feeds **two branches**: frame-wise classification + a **Boundary Regression Branch**; predicted boundaries refine/split the segmentation (+13.7% edit, +16.1% segmental F1 over MS-TCN-era SOTA on 50Salads/GTEA/Breakfast). **Why it fits:** the boundary head is exactly the missing "cut here" mechanism. **? Two caveats:** (1) ASRF's headline goal is to *reduce* over-segmentation (merge spurious fragments) ? the opposite of our over-*merge*; only the *split-at-a-detected-boundary* sub-mechanism transfers. (2) The claim that ASRF's boundary branch is a **model-independent post-hoc splitter** you can bolt onto any dense signal was **REFUTED** (§5.5) ? you must train the two-branch model, not graft the branch. **Effort/payoff:**

research bet.

- **TriDet** (Shi et al., CVPR 2023, [arXiv:2303.07347](https://arxiv.org/abs/2303.07347)). A **Trident-head** models each boundary as a *relative probability distribution over local bins* (boundary = expected value over neighboring instants) ? built for imprecise/ambiguous boundaries. 69.3% avg mAP on THUMOS14 (> ActionFormer 66.8% at ~75% latency), **convolution-only (SGP), no self-attention** (attractive on a single local GPU). Ablation shows the Trident-head's gain concentrates at strict tIoU=0.7 ? it buys *boundary precision*. **Effort/payoff:** research bet (trained on 100s of videos); the portable piece is the relative-boundary idea.

- **ActionFormer** (ECCV 2022, [arXiv:2202.07925](https://arxiv.org/abs/2202.07925)). The reference single-stage anchor-free template: multiscale pyramid + local self-attention; classify every moment + regress its boundaries (71.0% mAP@0.5 THUMOS14). This is the family TriDet/TemporalMaxer/ CausalTAD extend, and the natural target if we ever graduate from heuristic+LogReg. **Research bet.**

- **TemporalMaxer** (CVPRW 2023, [arXiv:2303.09055](https://arxiv.org/abs/2303.09055)) ? **the most important small-data signal in this dimension.** It replaces ActionFormer's self-attention block with a **parameter-free max-pool** and *beats* it (67.7 vs 66.8 avg mAP) at **~4× fewer params**. The ablation is the takeaway: a **learned 1-D conv block is the WORST** (59.4, overfits redundant features); parameter-free pooling wins. **Why it fits:** direct evidence to **keep our scorer parameter-light** (the numpy LogisticRegression over a handful of features is the *right* choice at our scale, not a placeholder to rush past). **Effort/payoff:** a discipline (free) ? adopt the principle, not the model. *(Caveat: the overfitting evidence is about feature redundancy on a mid-size set, a directional inference to our small-n setting.)*

- **CausalTAD** (2024, [arXiv:2407.17792](https://arxiv.org/abs/2407.17792)) ? *steering-away note.* Causal-attention masking (attend only to past/present) won EPIC-Kitchens/Ego4D 2024 tracks, but it adds **no** boundary-split module. Useful later for *online/streaming* rally detection; **NOT** the over-merge fix. Listed so we don't mistake causal attention for the remedy.

- **Metrics (the actionable cross-cut):** the field leads with **segmental Edit (Levenshtein) score and F1@k**, not frame accuracy (MoF) or plain tIoU ? see §5.4.

### 5.2 Shuttle-trajectory ? rally signal processing ? published badminton systems already do what we lack
This is the highest-confidence, lowest-effort dimension: in-domain papers operationalize the exact boundary heuristics we're missing, in spirit license-compatible (re-confirm before vendoring code).

- **See et al. ? Service & End-of-Rally Detection in Badminton** (2024 Springer chapter
[10.1007/978-3-031-69073-0_15](https://link.springer.com/chapter/10.1007/978-3-031-69073-0_15);

2025 SN Computer Science follow-on [10.1007/s42979-025-03935-0](https://link.springer.com/article/10.1007/s42979-025-03935-0)). **(a) Rally END = a sustained-inactivity rule:** the rally ends when **?80% of consecutive frames show no shuttle motion** (~**95% accuracy** over 187 rallies). **(b) Rally START (service) = a learned classifier** over court geometry + player poses (~26 features/frame × 12 frames ? CNN/BiLSTM, **88.6%** accuracy). **Why it fits:** the inactivity rule is *precisely* the end-of-rally cut our windowing lacks ? a few lines, directly breaks 65 s mega-windows. **? Caveats:** the rule is a *termination*, not a *within-rally split*; results rest on paywalled abstracts + small test sets (35?187 rallies); the serve classifier needs the (default-OFF) person cue + court polygons. **Effort/payoff:** the inactivity rule = **low-effort high-confidence win**; serve classifier = medium.

- **Hsu, Yu & Cheng ? trajectory + swing fusion** (Sensors 2024, 24(13):4372, [10.3390/s24134372](https://www.mdpi.com/1424-8220/24/13/4372)). Fusing TrackNet trajectory with a YOLOv7 swing detector via a Shot Refinement Algorithm raises hit detection **F1 72.3% ? 90.5%** (over 69 BWF matches / 1582 rallies). Hits are found by **trajectory direction-reversal / peak detection** (convert (x,y)?(frame,y) wave peaks) then reconciled with the swing cue. **Why it fits:** validates our **late/score-level fusion** approach in-domain, and direction-reversal/peak is a cheap **per-window feature** marking contact events (a "rally has hit cadence" signal ? the discriminator a "shuttle-moving" heuristic lacks). **? Important caveat (the one 2-1 vote):** this paper does *within-rally stroke/hit* segmentation, **not** rally start/end ? "it splits merged rallies" is *our extrapolation*; frame it as "hit cadence plausibly helps locate rally boundaries," not a proven rally-splitter. **Effort/payoff:** the peak/direction-reversal feature = low-effort win (feed the scorer); full swing fusion = medium.

- **What best separates "rally" from "shuttle merely moving"** (synthesis): sustained-inactivity termination (See), hit cadence / direction-reversals (Hsu), net-crossings ?2 (our Gate-A, [`rally_gate.py`](../backend/pipeline/detectors/rally_gate.py)), two-sided court occupancy (our person features). None alone is sufficient; the spine (§4) says **feed them all to the learned scorer**.

### 5.3 Small-data strategies ? most quality per label
- **Timestamp / sparse supervision** (Li et al., CVPR 2021, [arXiv:2103.06669](https://arxiv.org/abs/2103.06669); TAS survey Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352)). Label **one frame per action** ? iteratively refine pseudo frame-labels ? *comparable* to full supervision (50Salads: timestamp 75.6% acc / 60.1 F1@50 vs full 83.7% / 70.1). **Why it fits:** sparse per-rally timestamps cost far less than dense boundary labels ? relevant if we ever train a frame-wise model. **? Caveats:** "comparable" hides a real **~8?10 pt residual gap**; annotation savings

are modest (~35%, annotators still watch the whole clip); benchmarks are cooking/egocentric ? transfer to 1-D shuttle trajectories is **unproven**. **Effort/payoff:** research bet to pilot.
- **Parameter-light scorer** ? see TemporalMaxer (§5.1): keep the numpy LogisticRegression; resist a learned temporal net until the corpus is much larger. **Low-effort win (a discipline).**
- **? OPEN (the research could NOT verify these ? honest gap):** *active learning* (which video/segment to label next via uncertainty/diversity/core-set) and the *teacher-student / distillation governance boundary* for using Gemini as a weak/pseudo-labeler returned **no surviving verified claim**. Our pragmatic, guardrail-safe stance (not from the literature ? from our own design): pick the next video to label by **maximal local-vs-Gemini disagreement** (the §4.4 shadow signal) + **scorer uncertainty** + **venue/fps diversity**; and treat **Gemini strictly as an inference-time weak-reference / eval oracle ? never a training-data source for owned weights** (§8). These two topics deserve their own grounded research pass before we lean on them.

### 5.4 Evaluation rigor at small n ? lead with segmental metrics
- **Frame accuracy (MoF) and plain tIoU are BLIND to over-segmentation** (TAS survey, Ding et al., [arXiv:2210.10352](https://arxiv.org/pdf/2210.10352); segmental-F1 origin Lea et al., CVPR 2017). "The score can be high even when segments are fragmented." The over-merge that bit us is exactly what tIoU-F1 hides (a giant window still *overlaps* real rallies). **Action:** lead the leaderboard with the **segmental Edit (Levenshtein) score** (penalizes spurious ordering/fragmentation) **and F1@{0.1,0.25,0.5}**, paired with our existing `segment_count_ratio` + the merge/split breakdown. **We already have** F1@k, mAP@tIoU, segment-count-ratio, bootstrap CIs, paired sign-test, Platt/ isotonic + ECE/Brier ? the only *missing* primary metric is the **Edit score** (low-effort add).
- **? OPEN (not covered by a verified claim):** IAA / label-noise level in the golden set, and which over-segmentation-robust loss (ASRF boundary/smoothing, MS-TCN truncated-MSE) best tolerates it at small n. Measure IAA as the corpus grows (Roora has an A/A overlap plan); revisit robust losses only when a frame-wise model is on the table.

### 5.5 Refuted claims (recorded so we do NOT repeat them)
1. **"ASRF's boundary branch is model-independent ? a post-hoc splitter applicable to any dense segmentation output."** **REFUTED (0-3).** Implication: we cannot bolt ASRF's boundary head onto our trajectory signal as a drop-in; the two-branch model must be trained together. ? port the *concept*, don't expect a plug-in.
2. **"A GFL-inspired Boundary Distribution Regression head is the mechanism most relevant to splitting merged windows."** **REFUTED (0-3)** (and the cited source had a suspect identifier). Do not adopt this framing as established.

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## 6. Prioritized roadmap (cheap ? durable)

Every item is **its own measured PR**: default-OFF, scored on the §4 metric set over the golden set with honest stats (CI + sign-test), **re-confirmed on the served windowing** (the eval?serve gate), logged in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md). Ordered by the research's effort/confidence tiering. Each carries a **falsifiable success criterion** ? if it isn't met, the item does not promote (and that null is itself recorded).

### Definition of "bridging the gap"
Local rally segmentation, measured **honestly LOVO-by-match on the golden set**, approaches the Gemini reference (today **0.93 single-court / 0.66 multi-court** tIoU-F1) ? but tracked primarily through the **over-segmentation-sensitive metrics** (segmental Edit + F1@k + low merge/split rate), since those, not bare tIoU-F1, capture our actual failure. A secondary, free signal is rising **agreement-with-Gemini** on the unlabeled pool (§4.4, shadow only).

### Tier 0 ? Make the failure measurable FIRST (do before any fix)
- **E1 · Over-segmentation-aware leaderboard.** Add the **segmental Edit (Levenshtein) score** to `segmentation_metrics.py`; surface Edit + F1@{0.1,0.25,0.5} + `segment_count_ratio` + a **merge-rate / split-rate** breakdown (promote the `scratch/compare_local_vs_gemini.py` matching into a reusable `backend/eval/` helper) in the default `experiment.compare()` report; add the one-command rally-segmentation eval entrypoint (§4.5). **Success:** the current heuristic's over-merge shows up as a *number* (low Edit, high merge-rate) on the 6 golden videos, and a baseline row lands in the leaderboard. *(Low effort, high confidence ? research-backed §5.4.)*

### Tier 1 ? Low-effort, high-confidence wins
- **W1 · Bounded windowing (the direct over-merge fix).** Add a `max_window_duration` cap and a **sustained-inactivity split** to `trajectory_to_action_windows` ? port See et al.'s rule (split / end a window when ?~80% of consecutive frames over a trailing window show no shuttle motion). Config-gated, default preserves current behavior. **Success:** segmental Edit + F1@k improve vs `CONTROL` on the golden set (?F1 CI excludes 0 **and** sign-test agrees) with **no served-windowing regression**; merge-rate drops materially. *(Win ? §5.2 See et al., ~95% end-of-rally acc.)*
- **W2 · Rally-state features into the learned scorer (no special-casing).** Wire the already-built, default-OFF cues as scorer feature columns: **net-crossings** (Gate-A), **two-sided motion**, and a **new cheap trajectory feature ? direction-reversal / peak count** (Hsu) as a hit-cadence proxy; audio hit-rate when the hit detector lands. **Re-measure with W1 in place and on the served windowing** ? recall person was ?0.021 and audio +0.079 but *both CIs spanned 0* at n=6 on the *pre-fix over-merged* candidates; the picture may change once windows are bounded. **Success:** the ablation (`ablation.py`) shows a cue with ?F1 CI clearing 0 + sign-test, surviving served re-check

? promote that cue; others stay default-OFF + retained (no deletion on a small-n null). *(Win/medium.)*
- **W3 · Keep the scorer parameter-light (a discipline, documented not coded).** Per TemporalMaxer's ablation (learned temporal conv overfits; parameter-free wins), resist replacing the numpy LogisticRegression with a learned temporal net until the corpus is much larger. Record this as a standing decision. *(Win ? §5.1 TemporalMaxer.)*

### Tier 2 ? Medium effort
- **M1 · Serve-start classifier ? rally-start boundary** (See et al. pose/court, 88.6%). Needs the person cue + a **court-mask source** (manual polygon vs auto-homography ? *owner-gated* per [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) §9.1). **Success:** start-boundary timing error drops on golden; promotes only if it lifts segmental F1@k without served regression.
- **M2 · Per-venue threshold calibration** against golden labels (the heuristic's footage-fragility ? it scored 0.157 on multi-court). Tune `velocity_threshold` + `merge_gap` so local segment counts track golden counts per venue. **Success:** multi-court F1 rises toward the Gen-0 head's 0.561 without harming clean-court F1. *(Velocity is already fps-normalized ? calibrate per-venue, not per-fps.)*

### Tier 3 ? Research bets (durable; corpus-gated)
- **R1 · Owned segmenter with an explicit boundary branch** (the ASRF/TriDet *idea* ? a boundary head that predicts cut points ? ported into our owned TAL head, NOT the full deep model, and **trained**, not grafted (§5.5 refutation)). Per [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md): **TAL head first, VLM later.** **Gate:** the central unverified assumption is whether boundary-head gains hold at our scale on trajectory features ? so **pilot only when the corpus is materially larger (~?16?30 matches)**, and only after W1/W2 establish the heuristic ceiling. **Success:** beats the tuned heuristic + LogReg on held-out matches (segmental Edit + F1@k), honestly LOVO.
- **R2 · Timestamp/sparse supervision** to get more labeled rallies per annotator-hour (§5.3) ? pilot if/when we train a frame-wise model; research bet (residual gap, unproven on trajectories).

### Throughout ? corpus growth + active learning + label quality
- **Grow the golden set 6 ? ~16** via Roora ([`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md)) ? the single highest-leverage lever ([`NEXT_STEPS.md`](NEXT_STEPS.md)).
- **Active learning:** select the next videos to label by max **local-vs-Gemini disagreement** (§4.4) + scorer uncertainty + venue/fps diversity (our pragmatic stance; §5.3 flags the formal methods as an open research pass).
- **Measure IAA / label noise** as the corpus grows (Roora A/A overlap); revisit over-segmentation-robust losses only when a frame-wise model is on the table (§5.4 open question).

### Dead-ends / do-NOT (so we don't relitigate)
- Don't treat **causal attention** (CausalTAD) as the over-merge fix ? it has no boundary-split (§5.1).

- Don't expect **ASRF's boundary branch as a post-hoc drop-in** on our dense signal ? refuted (§5.5).
- Don't adopt a **learned temporal net** at n=6?16 ? TemporalMaxer evidence says it overfits (§5.1).
- Don't chase **bare tIoU-F1 / frame accuracy** ? blind to the over-merge (§5.4).
- Don't let **Gemini outputs become training data** for owned weights ? inference/shadow only (§8).

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## 7. Project: plan, tracking & definition of done

This roadmap is run as **one tracked project ? "Bridge the local?Gemini rally-detection gap."** It
**completes when all committed tiers land AND a single quality number is attributed to the effort**,
measured honestly and logged. This §7 is the living burn-down (update the status column as PRs merge;
log each result in [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md), numbers in
`eval_baselines/experiment_leaderboard.json`).

### 7.1 The number (what we attribute to this effort)
Measured **leave-one-video-out, grouped by match, on the golden set** (grown to ?~16):
- **Headline:** multi-court rally-segmentation **tIoU-F1**, `baseline ? final`, plus **gap-closure %**
  = `(final ? baseline) / (Gemini ? baseline)`. Today: heuristic **0.157** ? Gemini **0.66**
  (Gen-0 head already **0.561**). **Success bar (proposed, owner-adjustable): close ? 50% of the
  heuristic?Gemini multi-court gap** ? i.e. local multi-court F1 ? ~0.41, ideally beating Gen-0's 0.561.
- **Primary lens (over-segmentation-sensitive, the actual failure):** segmental **F1@0.25** +
  **Edit-score** + **merge-rate** ? these must improve even if tIoU-F1 moves little.
- **Reported with honest stats** (bootstrap CI + paired sign-test); a result counts only if it also
  holds on the **served** windowing (eval?serve gate). The **baseline is frozen at E1** before any fix.
- **Tracked headline = the n=6 golden *aggregate* tIoU-F1@0.5** (LOVO; the robust unit, not a single
  video) via `python -m backend.eval.rally_seg_eval`. The 0.157/0.66 figures above are per-venue
  *references*; computing an exact "% gap-to-Gemini closed" needs Gemini measured on the *same* golden
  set + metric ? **TODO** (don't mix the per-venue refs with the aggregate).

**Progress (cumulative, golden n=6 aggregate tIoU-F1@0.5):**

| after | F1 | merge_rate | ? vs baseline |
|---|---|---|---|
| E1 baseline (current heuristic) | 0.300 | 0.523 | ? |
| + W1 bounded windowing (cap=12, **default-OFF**) | **0.439** | **0.038** | **+0.139** (6/0 sign-test) |
| + W2 net-crossings gate (mc=1, **default-OFF**) | **0.457** | 0.038 | **+0.157 cumulative** (W2 step +0.019, 6/0) |
| + W2 multi-feature scorer (person/audio) | _next (GPU)_ | | |

### 7.2 Work items (status checklist)
Status: ? not started · ? in progress · ? done. (Each row = its own default-OFF, measured PR.)

| ID | Deliverable | Tier | Depends on | Acceptance / promote criterion | Status |
|---|---|---|---|---|---|
| **E1** | merge/split-rate metric + F1@k surfaced + one-command `rally_seg_eval` entrypoint; **froze baseline** (#185) | 0 | ? | over-merge shows up as a number on the 6 golden

videos | ? baseline F1 0.300, merge_rate 0.523 |
| **W1** | Bounded windowing ? `max_window_duration` cap + inactivity split (See et al.) (#187) | 1 | E1 | seg F1@k ? vs baseline (CI?0 **and** sign-test), no served regression; merge-rate ? | ? **?F1 +0.139 (6/0), merge_rate 0.523?0.038** (default-OFF) |
| **W2** | Rally-state features (net-crossings ? #188 + **never-empty safeguard** ?; two-sided motion, **direction-reversal/peak**, audio = follow-on) | 1 | E1, W1 | ?1 cue ?F1 CI clears 0 + sign-test, survives served re-check ? promote; others retained default-OFF | ? **net-crossings ?F1 +0.019 (6/0)**, gate now **do-no-harm** (never zeroes a video); multi-feature scorer next (GPU) |
| **W3** | Keep scorer parameter-light (standing decision; TemporalMaxer evidence) | 1 | ? | documented; no learned temporal net until corpus ? now | ? |
| **M1** | Serve-start classifier ? rally-start boundary (pose/court) | 2 | W1, **court-mask decision (owner-gated)** | start-boundary timing error ?; promote on seg-F1 lift, no served regression | ? |
| **M2** | Per-venue threshold calibration vs golden | 2 | E1 | multi-court F1 ? toward Gen-0 0.561 without harming clean-court | ? |
| **R1** | Owned segmenter with an explicit **boundary head** (port ASRF/TriDet *idea*, trained ? not grafted) | 3 (stretch) | corpus ?~16?30, W1/W2 | beats tuned heuristic+LogReg on held-out matches (seg Edit + F1@k), LOVO | ? |
| **R2** | Timestamp/sparse-supervision pilot (label-efficiency) | 3 (stretch) | R1 path | measured label-efficiency win | ? |
| **C** | Golden corpus 6 ? ~16 (Roora) + active-learning selection + measure IAA | ongoing | ? | corpus ?16; IAA recorded | ? (~6 now; ~10 by the weekend) |

### 7.3 Sequencing
`E1` (metrics + baseline) ? `W1`/`W2`/`W3` (the high-confidence wins) ? `M1`/`M2` ? `R1`/`R2`
(corpus-gated). `C` (corpus growth) runs in parallel throughout ? it is the single highest-leverage
input and gates the Tier-3 bets.

### 7.4 Definition of done
The project is **complete** when:
1. **Tier 0 + Tier 1 + Tier 2** work items are merged ? each a measured PR (default-OFF, promoted only
   on measured lift, re-confirmed on the served windowing).
2. **The headline number (§7.1) is recorded** in `QUALITY_ITERATIONS.md` + the leaderboard, with honest
   stats, on the grown golden set ? **and the gap-closure target is met, or the honest null is recorded
   with a reasoned go/no-go on Tier 3.**
3. Each landed item left the harness greener (no regression) and `QUALITY_ITERATIONS.md` tells the story.

> **Tier-3 scope decision (owner-adjustable):** R1/R2 (the owned boundary-head model + sparse
> supervision) are **corpus-gated stretch goals** ? the verified research says deep TAL detectors risk
> overfitting at n=6?16. **Default:** the project *completes* at Tier 0?2 + an attributed number, and
> Tier 3 spins out into the owned-model milestone ([`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md)).
> Fold Tier 3 into the DoD instead if you want the project to span the owned model.

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## 8. Constraints & guardrails (non-negotiable) + cross-links

- **Licensing:** every dependency ? code, data, AND weights ? must be **MIT / Apache-2.0 / BSD**.
  Reject viral/NC/custom (Gemma-3, Llama-4 MAU cap, Commons-Clause).
- **Never train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs** (terms/copyright);
  paid LLMs are inference/serving/shadow-reference only. Also forbidden: public datasets known to

be LLM-generated. **Exclude minors** from any training pool; per-user data needs explicit opt-in.
- **Owned model:** TAL head FIRST (de-risked, trains in minutes, doubles as a pseudo-label teacher +
honest baseline the VLM must beat), then Qwen3-VL (Apache-2.0) / InternVL3 (MIT) ? see
[`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).
- **Cross-links:** [`HOW_RALLY_DETECTION_WORKS.md`](HOW_RALLY_DETECTION_WORKS.md) ·
[`EXPERIMENT_HARNESS.md`](EXPERIMENT_HARNESS.md) · [`MULTI_SIGNAL_FUSION_PLAN.md`](MULTI_SIGNAL_FUSION_PLAN.md) ·
[`ANALYZERS_AND_RECONCILERS.md`](ANALYZERS_AND_RECONCILERS.md) · [`ABLATION_LAYER.md`](ABLATION_LAYER.md) ·
[`EVALUATION.md`](EVALUATION.md) · [`QUALITY_ITERATIONS.md`](QUALITY_ITERATIONS.md) ·
[`GOLDEN_SET_VENDORING.md`](GOLDEN_SET_VENDORING.md) · [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md).

#### 12. assistant (2026-06-16T15:15:01.058Z)

The cited file `output/local_vs_gemini_6.json` doesn't exist. The numbers in the doc are *aggregate/per-venue* references, not per-video ground-truth scores for mahadevpura-2. Let me check the leaderboard and the QUALITY_ITERATIONS log for the actual recorded per-video numbers, especially the mahadevpura-2 hard-cap F1@0.25 0.722 / mIoU 0.50 and the count figures (n=40, count_ratio 1.03, mean 13.8s, max 43.3s).

#### 13. user (2026-06-16T15:15:02.477Z)


```
{
  "name": "gemini_wrapper_baseline",
  "description": "Commodity-VLM wrapper baseline: the full pipeline's `gemini` segmenter scored vs human golden labels at IoU 0.5. The commoditization/moat reference the owned model must be judged against ? on multi-court footage the wrapper drops 0.93 -> 0.66, so the owned model's target is the multi-court number, not the clean-footage one.",
  "model": "gemini-flash-latest (resolved 2026-06-10; 3.5-flash era)",
  "conditions": "serial uploads (ai_max_workers=1), chunks re-encoded to 1280px w/ audio (ai_chunk_scale_width), 6-attempt exponential generate retry; min_segment_duration=3.0 filter active (rejects real ~2.5s rallies ? see notes)",
  "per_video": {
    "mahadevpura_singles": {
      "f1": 0.93, "precision": 1.00, "recall": 0.87, "mean_iou": 0.80,
      "boundary_err_s": 0.7, "n_gt": 31, "n_pred": 27,
      "notes": "3 of 4 FNs were detected by Gemini but rejected by min_segment_duration=3.0 (2.5s rallies). Chunk 6 (unlabeled tail) lost to 503s ? no effect on the labeled window."
    },
    "kushagra_singles": {
      "f1": 0.66, "precision": 0.69, "recall": 0.62, "mean_iou": 0.73,
      "boundary_err_s": 1.2, "n_gt": 32, "n_pred": 29,
      "notes": "Multi-court (contrast 1.4x). Full 7/7 chunk coverage, audio intact. FPs cluster as background-game pickups + one 53s dead-time merge ? the exact confound the contrast metric predicts."
    }
  },
  "comparison_at_time_of_run": {
    "heuristic_lovo": {"mahadevpura_singles": 0.657, "kushagra_singles": 0.157},
    "gen0_learned_lovo": {"mahadevpura_singles": 0.744, "kushagra_singles": 0.561}
  },
  "reliability_note": "gemini-flash-latest served 503 capacity waves throughout (one chunk needed a ~5min backoff); a sole-provider dependency is operationally flaky ? the provider-registry fallback design is load-bearing.",
  "cost_note": "well under $1 for both videos at flash pricing (serial, ~14 chunk requests total)."
}
```

### 14. user (2026-06-16T15:15:03.967Z)

```
Found 34 files
docs\RALLY_QUALITY_RESEARCH.md
docs\QUALITY_ITERATIONS.md
tests\test_rally_seg_eval.py
backend\eval\rally_seg_eval.py
tests\test_promotion.py
tests\test_segmentation_metrics.py
tests\test_served_gate_a_regression.py
tests\test_eval_metrics.py
tests\test_experiment_honest.py
tests\test_eval_batch.py
eval_baselines\fixtures\multi_rally_with_deadtime\expected.json
eval_baselines\fixtures\occlusion_break\expected.json
eval_baselines\gemini_wrapper_2026-06-10.json
eval_baselines\fixtures\clean_single_rally\expected.json
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md
docs\archives\quality-iterations\2026-06-human-gt-tuning\README.md
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md
eval_baselines\experiment_leaderboard.json
docs\CODE_MAP.md
docs\ANALYZERS_AND_RECONCILERS.md
backend\eval\golden_fixtures.py
backend\eval\harness.py
backend\eval\metrics.py
backend\eval\segmentation_metrics.py
backend\eval\served_gate_a.py
backend\eval\ablation.py
backend\eval\batch.py
backend\eval\calibrate_wasb.py
backend\eval\calibration.py
backend\eval\eval_partial_labels.py
backend\eval\experiment.py
backend\eval\export_wasb_segments.py
backend\eval\fusion_compare.py
```

### 15. assistant (2026-06-16T15:15:08.858Z)

```
The Gemini wrapper baseline shows mahadevpura_singles with n_gt 31 (not 39 or 40) ? but that's a
different clip/labeling than "mahadevpura-2". Let me search QUALITY_ITERATIONS.md and the
leaderboard for the specific hard-cap numbers cited in the claim.
```

### 16. user (2026-06-16T15:15:11.007Z)

```
72-| tIoU-F1@0.5 | 0.439 | **0.457** | **+0.019** (95% CI [+0.011,+0.027], **sign-test 6/0**,
p=0.031) |
73-| F1@0.25 | 0.772 | 0.805 | +0.033 (6/0) |
74:| segment_count_ratio | 1.44 | 1.33 | over-split trimmed |
75-
76-Small but **statistically robust** (every video improves) and **served-measured** ? and a
satisfying reversal of the
--
97-|---|---|---|---|
98-| tIoU-F1@0.5 | 0.300 | **0.439** | **+0.139** (95% CI clears 0; **sign-test 6/0**, p=0.031)
|
99:| F1@0.25 | 0.501 | **0.772** | +0.271 |
100-| merge_rate | 0.523 | **0.038** | ?0.485 (over-merge essentially eliminated) |
101:| segment_count_ratio | 0.700 | 1.44 | now mildly over-*splits* ? W2's scorer to clean up |
102-
103-**Robust, not overfit:** every cap in 6?20 s clears 0 with 6/0 sign-test (the win isn't a
knife-edge); smaller caps trade
--
128-|---|---|
```

```
129-| tIoU-F1@0.5 | **0.300** [0.119, 0.494] |
130:| F1@0.25 | 0.501 [0.239, 0.747] |
131-| **merge_rate** | **0.523** [0.258, 0.788] |
132:| segment_count_ratio | 0.700 [0.405, 0.957] |
133-
134-The `0.300` **matches the independent served-side measurement below** (min_crossings=0 =
`wasb_hybrid` = 0.300) ? cross-validated.
--
323-  match) so a *learned* producer's outputs used as arbiter features are out-of-fold (no
leakage).
324-- **C4 ? over-segmentation-visible metrics.** `backend/eval/segmentation_metrics.py`:
`f1_at_overlaps`
325:  (F1@{10,25,50}), `average_precision`/`mean_average_precision` (mAP@tIoU),
`segment_count_ratio`. tIoU alone
326-  is blind to a rally split in 3 or two rallies merged ? exactly what the reconciler exists
to fix.
327-
```

#### 17. assistant (2026-06-16T15:15:15.794Z)

```
The QUALITY_ITERATIONS.md numbers are the n=6 golden *aggregate* (F1@0.5 0.300?0.439?0.457,
F1@0.25 0.501?0.772?0.805). None of these are the per-video mahadevpura-2 hard-cap numbers
(F1@0.25 0.722, mIoU 0.50) cited in the claim. Let me check the leaderboard JSON and search for
the specific mahadevpura-2 per-video breakdown and the W1 count figures (n=40, count_ratio 1.03,
mean 13.8s, max 43.3s).
```

#### 18. user (2026-06-16T15:15:17.623Z)

```
107-      "owner_approved": true,
108-      "real_video_corroboration": "GX010125_1080p offline re-score: 45 candidates -> CONTROL
45 rallies vs Gate-A 27 (dropped 18 low-crossing held-shuttle/feed windows, ~1.7min)",
109:      "over_seg_segment_count_ratio": {"before": 1.68, "after": 1.01},
110-      "source_archive": "docs/archives/quality-iterations/2026-06-first-ab-
experiment/README.md"
111-    },
```

#### 19. assistant (2026-06-16T15:15:21.253Z)

```
The leaderboard doesn't contain the mahadevpura-2 per-video numbers either. Let me do a broad
search for where the specific cited figures (0.722, mIoU 0.50, count_ratio 1.03, mean 13.8, max
43.3, the 40 hand-labels) actually live ? possibly in the artifacts/ dir (untracked, just
created) or a scratch report.
```

#### 20. user (2026-06-16T15:15:22.890Z)

```
---OUTPUT DIR---
total 292
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:34 .
drwxr-xr-x 1 avidu 197609      0 Jun 16 11:24 ..
drwxr-xr-x 1 avidu 197609      0 Jun 16 11:14 chunks
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 highlights_work
-rw-r--r-- 1 avidu 197609 212992 Jun 16 20:34 val.db
-rw-r--r-- 1 avidu 197609  61195 Jun 16 20:38 val_extract.log
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:38 val_reels
drwxr-xr-x 1 avidu 197609      0 Jun 16 20:34 wasb
---root json---
config.json
```

#### 21. user (2026-06-16T15:15:23.775Z)

```
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-584-**In-domain prior
art that already hard-codes our cues** (so our delta is the *no-branch discipline*, not the
docs\ANALYZERS_AND_RECONCILERS.md-585-cues): non-broadcast badminton rally start/end (See et al.
```

2024/2025), shuttle-state/hit detection (MonoTrack
docs\ANALYZERS_AND_RECONCILERS.md-586-arXiv:2204.01899; Hsu/Yu/Cheng, Sensors 2024 24(13):4372),
service-fault (Goh et al., Sensors 2023 23(24):9759),
docs\ANALYZERS_AND_RECONCILERS.md:587:tennis in-play state machines
(10.1080/19346182.2013.819007), badminton point-segmentation (Ghosh et al. 2018,
docs\ANALYZERS_AND_RECONCILERS.md-588-arXiv:1712.08714), ShuttleSet (arXiv:2306.04948).
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-589-
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-590-### 13.2
Terminology to adopt (gloss our nouns with the field's on first use)
--
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-657-  arbiter can't
represent AND/XOR; hand-crossed products recover it without leaving "signal not verdict."
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-658-  *(Snoek 2005.)* ?
**producers + scorer.**
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-659-- **C8 ? represent
a missing cue as an explicit `is_present`/uncertainty column, not a silent 0** ? "cue
docs\ANALYZERS_AND_RECONCILERS.md:660:  absent" ? "cue says no" for a linear arbiter. *(DBF
arXiv:1511.03183.)* ? **producers + scorer.**
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-661-- **C9 ? keep warm-
up/session-state a SOFT feature, never a hard pre-filter**, and add a court/context
docs\ANALYZERS_AND_RECONCILERS.md-662-  invariance probe: the static court co-occurs with
rallies, so the scorer can learn "court visible" instead
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\ANALYZERS_AND_RECONCILERS.md-663-  of "point in
play" (action-context confusion). Evaluate on warm-up stretches that share the court but lack
--
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md-47-**Per-video held-out
F1 (IoU 0.5):**
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-48-| video | CONTROL | Gate-A | learned (LOVO) |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-49-|---|---|---|---|
docs\archives\quality-iterations\2026-06-first-ab-experiment\README.md:50:| adarsh_avi_singles |
0.597 | **0.722** | 0.613 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-51-| kushagra_singles | 0.154 | **0.163** | 0.163 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-52-| largetest_doubles | 0.593 | **0.667** | 0.535 |
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-first-ab-
experiment\README.md-53-| gbaaddy | 0.231 | **0.274** | 0.222 |
--
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-
safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-42-TAS head, given ASFormer's
small-data result used 2048-d I3D features?* This session answered it.
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-43-
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-44-- **Cross-video LOVO of the current
heuristic** (`calibrate_local`, Adarsh-96 + Kushagra-32, human labels):
docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md:45:  mean held-out F1 **0.443 ± 0.279**;
Adarsh 0.722, **Kushagra 0.163** (its own ceiling ? *no* velocity-windowing
C:\Users\avidu\Projects\badminton-highlight-
indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-46-
config does better). **The ±0.279 spread is method variance, not config.** ? This also
**corrects the study's

docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-47-  "~0.73 bar"**: that was a fit-on-self single-match number; the honest cross-video heuristic bar is **~0.44**,
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\research\OWNED_MODEL_TRAINING_STUDY.md-48-which *lowers* the Gen-0 ship gate.
--
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-34-```
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-35-per-video best-fit: adarsh 0.777   |   kushagra 0.163  (ceiling ? NO config beats it)
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-36-leave-one-video-out: baseline mean 0.316 ? calibrated 0.443 ± 0.279  (overfit gap +0.027, tiny)
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:37:   held-out adarsh   0.722 (config tuned on Kushagra ? Adarsh's own optimum ? config TRANSFERS)
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-38-   held-out kushagra 0.163   (config tuned on Adarsh = Kushagra's own ceiling)
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-39-```
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-40-**The asymmetry is the finding.** Tuning generalizes fine *where the heuristic can solve the video*
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:41:(Kushagra?Adarsh = 0.722). But **Kushagra is unsegmentable by *any* velocity-windowing config (0.163)** ? and
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-42-this is NOT a WASB-recall failure: WASB tracks the shuttle **better** on Kushagra than Adarsh (**97% of GT
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-43-rallies contain ?2 net-crossings**, median 5, 310 visible pts\rally, vs Adarsh 76%). The likely mechanism:
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-44-Kushagra's shuttle is visible 67.5% of the time (vs Adarsh 40%) ? tracked *through* inter-rally activity ? the
--
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-145-### Learned prototype LOVO (n=5)
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-146-| held-out | per-frame AUC | learned F1 (LOVO) | oracle F1 | heuristic best |
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-147-|---|---|---|---|---|
docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md:148:| Adarsh (singles) | 0.879 | 0.709 | 0.722 | 0.73 |
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-149-| Kushagra (multi) | 0.853 | **0.561** | 0.633 | 0.16 |
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-150-| LargeTest (doubles) | 0.915 | 0.713 | 0.756 | 0.75 |
C:\Users\avidu\Projects\badminton-highlight-indexer\.claude\worktrees\w2-safeguard\docs\archives\quality-iterations\2026-06-multivideo-lovo\README.md-151-| gBaaddy (multi) | 0.833 | **0.574** | 0.574 | 0.28 |

### 22. assistant (2026-06-16T15:15:31.136Z)

This is an important finding. The cited per-video mahadevpura-2 numbers (F1@0.25 0.722, mIoU 0.50) do NOT appear anywhere in the tracked repo data. The only 0.722 figures that exist are for completely different videos (adarsh_avi_singles Gate-A 0.722; Adarsh held-out LOVO 0.722). The cited data file `output/local_vs_gemini_6.json` does not exist. Let me check the val.db and the val_extract.log in output/ since those are the most recent (Jun 16 20:38) and might contain the actual June-12 validation run data referenced in the claim.

**23. user (2026-06-16T15:15:33.293Z)**

```
['videos', 'play_segments', 'sqlite_sequence', 'events', 'candidate_segments', 'jobs',
'user_models']
```

**24. user (2026-06-16T15:15:35.280Z)**

```
No matches found
```

**25. user (2026-06-16T15:15:40.114Z)**

```
Traceback (most recent call last):
  File "<string>", line 5, in <module>
    for r in cur.execute('SELECT id, filename FROM videos').fetchall(): print(r)
             ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
sqlite3.OperationalError: no such column: filename
--- videos ---
```

**26. user (2026-06-16T15:15:44.368Z)**

```
videos cols: ['id', 'filepath', 'processed_at', 'status', 'validation_results', 'user_id']
('db84f3e65318e49aea8320de4c44f7d1', 'output\\highlights_work\\GX010128.MP4', '2026-06-16
12:32:39', 'processed', '[]', None)
('99a3e73afde3933e113767f2d66f71fe', 'output\\highlights_work\\GX010129.MP4', '2026-06-16
13:39:53', 'processed', '[]', None)
('6460c2fc7647e89537fc6afba80b519a', 'output\\highlights_work\\mahadevpura-0.MP4', '2026-06-16
14:20:31', 'processed', '[]', None)
--- play_segments by video ---
('6460c2fc7647e89537fc6afba80b519a', 62, 0.016683333333333335, 768.7513166666666)
('99a3e73afde3933e113767f2d66f71fe', 50, 0.016683333333333335, 536.7528833333333)
('db84f3e65318e49aea8320de4c44f7d1', 78, 0.016683333333333335, 755.70495)
```

**27. assistant (2026-06-16T15:15:51.257Z)**

This is very revealing. The val.db only contains 3 videos: GX010128 (78 segments), GX010129 (50), mahadevpura-0 (62). These match the "W1 cap=12" column counts in the prompt's table exactly (GX010128: 78, GX010129: 50, mahadevpura-0: 62). But mahadevpura-2 is NOT in this database at all.

So the actual local validation run that exists in the repo covered only 3 videos, and the W1-cap counts (78, 50, 62) correspond to those three. The mahadevpura-2 data (the entire ground-truth claim) is not present in any tracked or output file I can find. Let me confirm by searching the candidate_segments and the whole tree once more for the cited mahadevpura-2 ground-truth numbers (40 labels, mean 5.7s, F1@0.25 0.722, mIoU 0.50, boundary med 1.74/3.12).

**28. user (2026-06-16T15:15:53.392Z)**

```
Traceback (most recent call last):
  File "<string>", line 7, in <module>
    for r in cur.execute('SELECT video_id, count(*) FROM events GROUP BY video_id').fetchall():
print(r)
             ~~~~~~~~~~~^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
sqlite3.OperationalError: no such column: video_id
candidate_segments by video:
('6460c2fc7647e89537fc6afba80b519a', 62)
('99a3e73afde3933e113767f2d66f71fe', 50)
('db84f3e65318e49aea8320de4c44f7d1', 78)
events by video:
```

**29. user (2026-06-16T15:15:54.225Z)**

```
Found 6 files
docs\RALLY_QUALITY_RESEARCH.md
docs\archives\research\AUDIO_RALLY_DETECTION_PLAN.md
```

```
docs\archives\research\RALLY_DETECTION_RESEARCH.md
docs\PMF_MARKET_RESEARCH.md
docs\GOLDEN_SET_PHASE1_VIDEOS.md
docs\ANALYZERS_AND_RECONCILERS.md
```

### 30. assistant (2026-06-16T15:15:58.263Z)

Let me check these 6 files for any of the specific mahadevpura-2 ground-truth numbers, to be thorough before concluding the data isn't present.

### 31. user (2026-06-16T15:16:00.394Z)

```
110-- [Unstructured-video event detection ? arXiv 2002.08097](https://arxiv.org/pdf/2002.08097)
111-- [Badminton service-height CV setup (camera 4.5 m / 30°) ?
PMC](https://pmc.ncbi.nlm.nih.gov/articles/PMC10747833/)
112:- [VideoBadminton dataset ? arXiv 2403.12385](https://arxiv.org/html/2403.12385v1)
113-- [Table-tennis frame-rate vs detection accuracy ? Nature Sci
Rep](https://www.nature.com/articles/s41598-023-42966-6)
114-- [OpenTTGames (120 fps) dataset](https://lab.osai.ai/)
```

### 32. user (2026-06-16T15:16:02.476Z)

```
5ba6955 fix(fusion): W2 net-crossings gate is do-no-harm ? never empty a non-empty video (#188
follow-up)
64494ef feat(eval): golden regression fixtures M1 ? video-free CONTROL-only characterization
harness (#189)
a471745 feat(eval): Tier 1 (W2) ? net-crossings rally-state gate on W1 (?F1 +0.019, 6/0) (#188)
1de2817 docs: freeze Golden Regression Fixtures plan + adopt tracked-project-doc template (#186)
a3b209a feat(windowing): Tier 1 (W1) ? bounded windowing fixes the over-merge (?F1 +0.139, 6/0)
(#187)
90ab087 feat(eval): Tier 0 (E1) ? over-segmentation-aware rally-seg eval + frozen baseline
(#185)
960c00d docs: rally-quality research + eval-framework roadmap (foundation doc) (#184)
b6a39c4 feat(runner): fully-local no-AI segmenter mode (+ WASB stream flag, .env load) (#183)
---branch---
feat/w2-never-empty-safeguard
---untracked/modified---
 M config.json
```

### 33. assistant (2026-06-16T15:16:14.446Z)

I now have a complete picture. Let me summarize what the data does and does not support.

The cited data file `output/local_vs_gemini_6.json` does not exist anywhere in this worktree. The only local validation database (`output/val.db`, the most recent run, Jun 16) contains exactly 3 videos: GX010128, GX010129, and mahadevpura-0 ? and crucially, mahadevpura-2 is NOT among them. The W1-cap counts in the val.db (78, 50, 62) match the prompt's "local W1 cap=12" column for exactly those 3 videos.

Every specific numeric pillar of the mahadevpura-2 claim is absent from all tracked/output data I can find:
- The hard-cap F1@0.25 0.722 / mean-IoU 0.50 / F1@0.5 0.296 figures appear nowhere. The only 0.722 values in the repo belong to entirely different videos (adarsh_avi_singles Gate-A 0.722; Adarsh LOVO held-out 0.722) ? a likely source of number-confusion.
- The W1 count figures for mahadevpura-2 (n=40, count_ratio 1.03, mean 13.8s, max 43.3s), the 40 owner hand-labels, mean 5.7s, and the boundary-error medians (1.74/3.12) do not appear in QUALITY_ITERATIONS.md, the leaderboard, the Gemini baseline, or val.db.
- The Gemini baseline that does exist (`gemini_wrapper_2026-06-10.json`) lists a "mahadevpura_singles" with n_gt 31, n_pred 27 ? different counts and a different clip than the "mahadevpura-2 / Gemini 39" cited.

Per the instruction "you MAY Read output/local_vs_gemini_6.json" and "ONLY the data," that data is not present, so the claim's evidence cannot be corroborated. I default to holds=false.

### 34. user (2026-06-16T15:16:26.090Z)

```
Structured output provided successfully
```

### 35. assistant (2026-06-16T15:16:32.187Z)

```
I have completed the adversarial verification and returned the result via StructuredOutput.

Summary of what I found: the claim does NOT hold against the available data. The cited evidence
file `output/local_vs_gemini_6.json` does not exist anywhere in the worktree, and every
distinctive number in the mahadevpura-2 claim (the W1 counts, the 40-label ground truth, and
especially the hard-cap F1@0.25 0.722 / mIoU 0.50) is absent from all tracked and output data.
The only local validation database present (`output/val.db`) covers a different 3-video set that
does not include mahadevpura-2, and the only 0.722 figures in the repo belong to unrelated
videos (adarsh / Adarsh LOVO).
```